in Figure 5 for the SA-APAC macro-region, results for the other regions are similar and are available upon request.

Comparing it with Figure 4, we notice that the incentive has no effect on the default decision, instead the region with a positive abatement expenditure enlarges: the yellow color in the last row of Figure 5 (dealing with u/y) corresponds to the region where a positive abatement expenditure is optimal if $p = 0.2$, but not in the $p = 0$ case. As far as consumption ratio is concerned, we observe an improvement only for a high output because for intermediate values the consumption rate is already high.

Considering Table 7, where optimal default and abatement expenditure are reported for each country, we observe that only Malaysia would invest in climate risk adaptation. Results not reported for sake of brevity show that the 20% incentive would also lead Thailand, China, as well as the median case, to invest in climate risk abatement. As expected, there is no change in the default decision.

We now assume that the probability of being readmitted to the market after the default - and to issue new debt - depends on the stock of pollution, that is:

$$\chi = \chi(f) = \chi_b e^{-\alpha_\chi f},$$

χ_b being the baseline parameter. In this case, increasing the pollution level, the value of χ, and so the probability of exiting from the autarky, decreases, approaching zero when the pollution is high. Therefore climate risk enters in the debt restructuring and in the readmission to the market and the country should have an incentive to invest in pollution abatement. This opportunity seems to have a negligible impact both on the default and the risk adaptation expenditure; the effect on the consumption ratio is mixed but still small, see Figure 6, where we set $\alpha_\chi = 0.5$. Results with $\alpha_\chi = 1$, therefore increasing the sensitivity of χ with respect to f, here not reported, show a similar behavior.

8 Conclusions

There is empirical evidence of a positive relationship between sovereign debt rates and climate risk but, up to now, no theoretical foundation for it has been established. The model presented in this paper aims to shed light on this relationship considering climate risk and endogenous default on public debt.

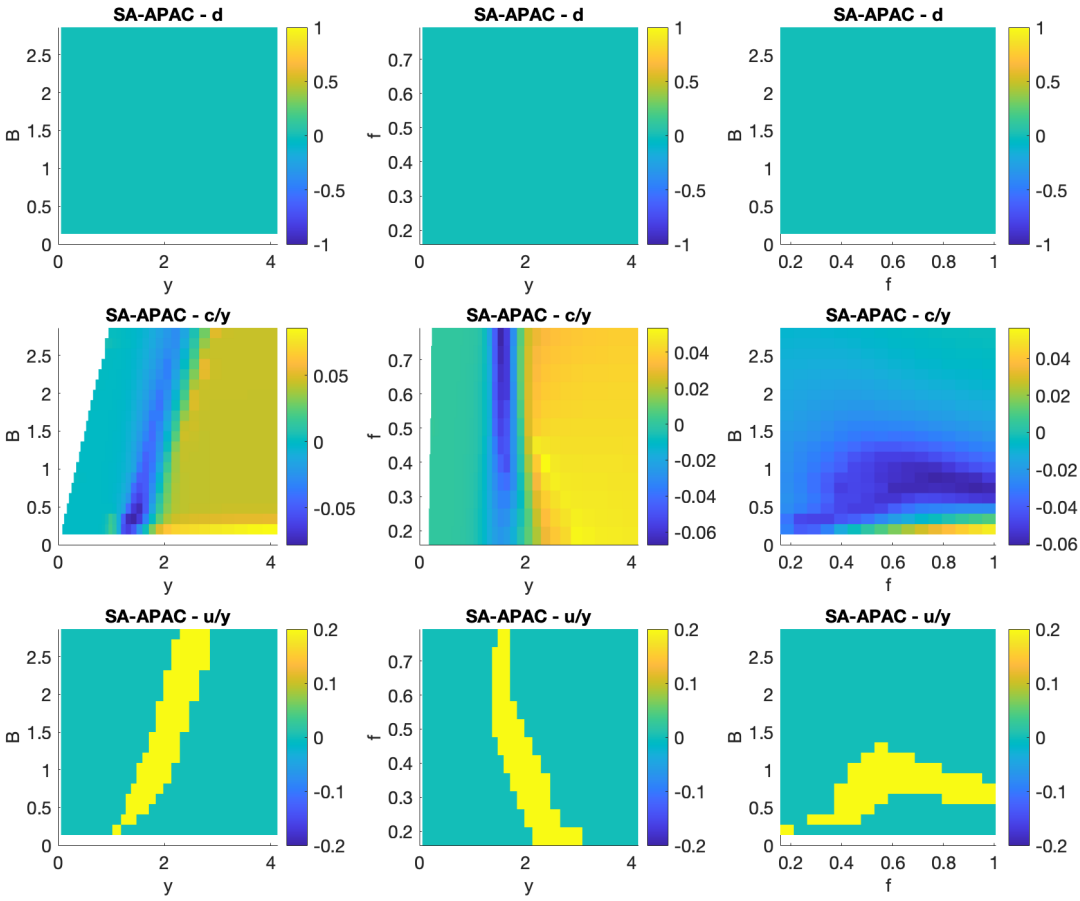

Figure 5: Carbon offset incentive. Differences of the optimal control values obtained considering the incentive and the baseline model, that is, dealing for example with d we show $d_{incentive} - d_{baseline}$ for the SA-APAC region, see Figure 4 for details on the baseline analysis.

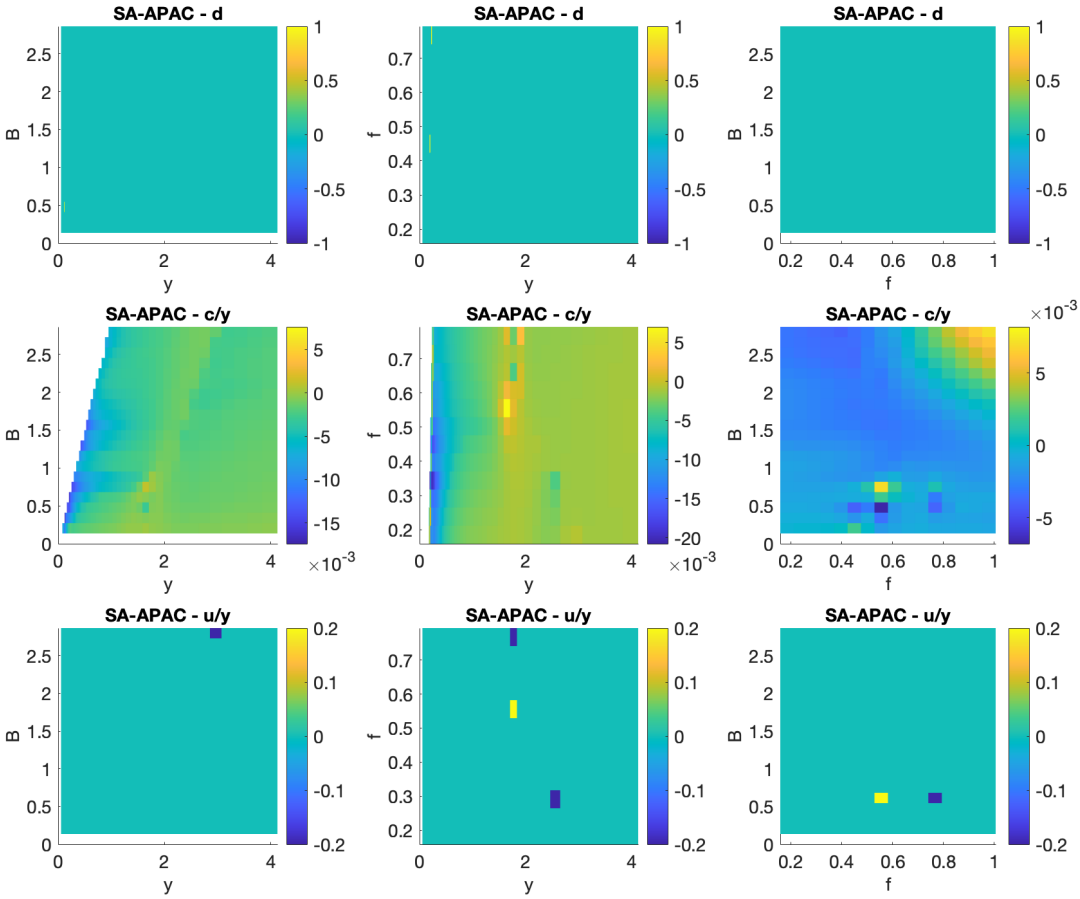

Figure 6: Readmission to the market dependent on the pollution stock. Differences of the optimal control values obtained considering the modified probability of readmission and the baseline model, that is, dealing for example with d we show $d_{modified} - d_{baseline}$ for the SA-APAC region, see Figure 4 for details on the baseline analysis.